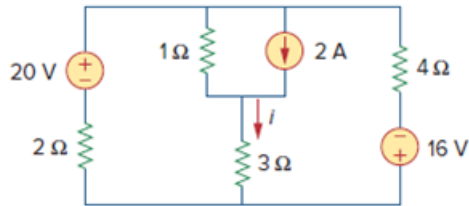


Problem

For the circuit in Fig. 4.83, use superposition to find i . Calculate the power delivered to the 3- Ω resistor.

Figure. 4.83:



Step-by-step solution

Step 1 of 11

Refer to Figure 4.83 in the textbook.

Apply Superposition principle to the circuit.

Consider a single source among three sources and obtain the values of current i_1 , i_2 , and i_3 individually.

$$i = i_1 + i_2 + i_3 \dots\dots (1)$$

Step 2 of 11

Consider the voltage source of 20 V and open circuit the current source of 2 A and short circuit the voltage source of 16 V as shown in Figure 1.

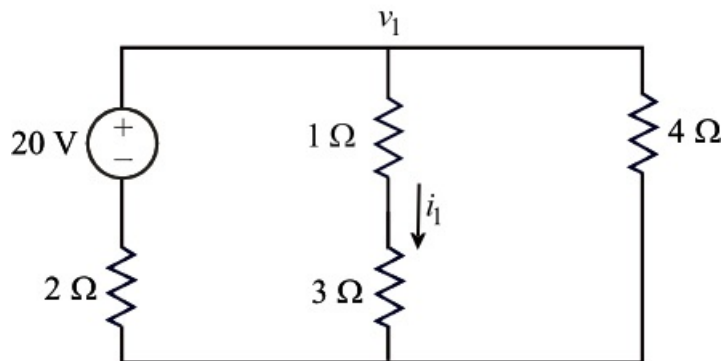


Figure 1

Step 3 of 11

Write the Kirchhoff's current law equation at node v_1 .

$$\frac{v_1 - 20}{2} + \frac{v_1}{1+3} + \frac{v_1}{4} = 0$$

$$\frac{2v_1 - 40 + v_1 + v_1}{4} = 0$$

$$4v_1 = 40$$

$$v_1 = 10 \text{ V}$$

From Figure 1, the current is,

$$i_1 = \frac{v_1}{1+3}$$

$$= \frac{10}{4}$$

$$= 2.5 \text{ A}$$

Step 4 of 11

Consider the current source of 2 A and short circuit the voltage sources of 16 V and 20 V as shown in Figure 2.

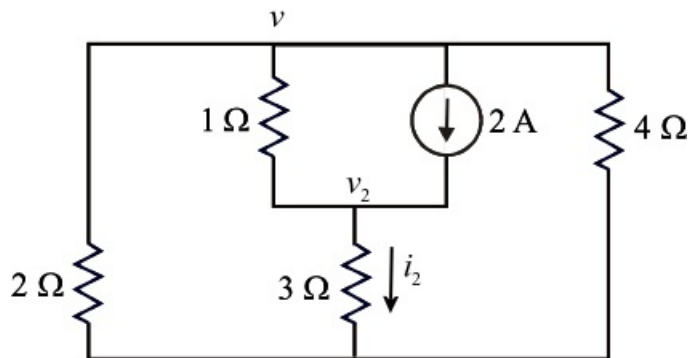


Figure 2

Step 5 of 11

Write the Kirchhoff's current law equation at node v .

$$\frac{v}{2} + \frac{v - v_2}{1} + \frac{v}{4} + 2 = 0$$

$$\frac{2v + 4v - 4v_2 + v + 8}{4} = 0$$

$$7v - 4v_2 = -8 \dots\dots (2)$$

Write the Kirchhoff's current law equation at node v_2 .

$$\frac{v_2}{3} + \frac{v_2 - v}{1} - 2 = 0$$

$$\frac{v_2 + 3v_2 - 3v - 6}{3} = 0$$

$$4v_2 - 3v - 6 = 0$$

$$v = \frac{4v_2 - 6}{3} \dots\dots (3)$$

Step 6 of 11

Substitute $\frac{4v_2 - 6}{3}$ for v in equation (2).

$$7\left(\frac{4v_2 - 6}{3}\right) - 4v_2 = -8$$

$$9.33v_2 - 14 - 4v_2 = -8$$

$$5.33v_2 = 6$$

$$v_2 = 1.125 \text{ V}$$

Step 7 of 11

From Figure 2, the current is,

$$\begin{aligned} i_2 &= \frac{v_2}{3} \\ &= \frac{1.125}{3} \\ &= 0.375 \text{ A} \end{aligned}$$

Step 8 of 11

Consider the voltage source of 16 V and open circuit the current source of 2 A and short circuit the voltage source of 20 V as shown in Figure 3.

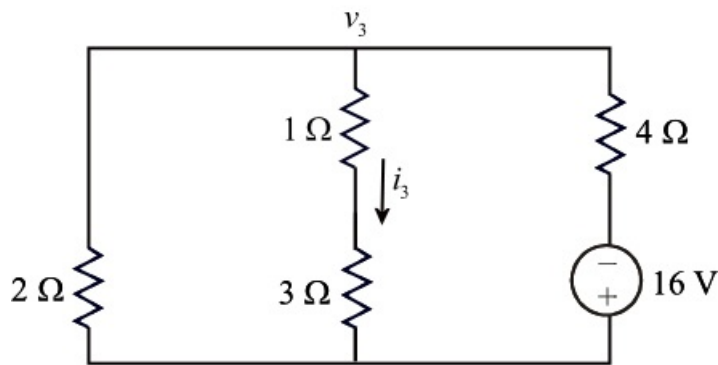


Figure 3

Step 9 of 11

Write the Kirchhoff's current law equation at node v_3 .

$$\frac{v_3}{2} + \frac{v_3}{1+3} + \frac{v_3+16}{4} = 0$$

$$\frac{2v_3 + v_3 + v_3 + 16}{4} = 0$$

$$4v_3 = -16$$

$$v_3 = -4 \text{ V}$$

From Figure 3, the current is,

$$\begin{aligned} i_3 &= \frac{v_3}{1+3} \\ &= \frac{-4}{4} \\ &= -1 \text{ A} \end{aligned}$$

Step 10 of 11

Recall equation (1).

$$i = i_1 + i_2 + i_3$$

Substitute 2.5 for i_1 , 0.375 for i_2 , and -1 for i_3 .

$$\begin{aligned} i &= 2.5 + 0.375 - 1 \\ &= 1.875 \end{aligned}$$

Therefore, the value of the current, i is 1.875 A.

Step 11 of 11

Calculate the power delivered to $3\ \Omega$ resistor.

$$\begin{aligned} P_{3\Omega} &= 3i^2 \\ &= 3(1.875)^2 \\ &= 10.547\ \text{W} \end{aligned}$$

Therefore, the power delivered to $3\ \Omega$ resistor, $P_{3\Omega}$ is 10.547 W.